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The University shall primarily provide advanced instruction and professional training in the arts, philosophy, social sciences, agriculture and fishery, forestry, science and technology engineering, education, law and other related fields. It shall also undertake research and extension services and provide progressive leadership in its areas of specialization.

Section 2, R.A. 9313

Our Vison

A transformative University committed to technology, innovation, and service excellence

Board Resolution No. 41, s. 2014

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CC101- INTRODUCTION TO COMPUTING

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College of Arts & Sciences

PREFACE

This module provides an overview of the Computing Industry and Computing Profession, including Research and Applications in different fields: and appreciation of computing in different fields such as Biology, Sociology, Environment and Gaming; an Understanding of ACM Requirements; an Appreciation on the History of Computing and Knowledge of the Key Components of Computer Systems (Organization and Architecture), Malware, Computer Security, Internet and Internet Protocols, HTML 4/5 and CSS.

UNIT I: Computer Careers and Certification
UNIT II: Digital Literacy: Introducing a World of Technology
UNIT III: Computer Systems (Organization and Architecture)
UNIT IV: Introduction to HTML
UNIT V: Introduction to Cascading Style Sheets

Each unit composed of six (6) parts namely: Learning Outcomes, Introduction, Discussion, Assessment, References, and Acknowledgement.

Learning Outcomes – In this part, students will identify what they will know and be able to do by the end of a **course** or program.

Introduction - This part describes the scope, and brief explanation or summary of the certain topic.

Discussion – In this part, students will learn and explore the different topics of the unit.

Assessment – In this section, students need to answer several questions related to the topics discussed in order for the instructor to evaluate their learnings.

References – This part presents the citations of sources of information.

Acknowledgement – In this section, references/authors are recognized.

UNIT 1
Computer Careers and Certification

1.0 Learning Outcomes

After completing this module, you will be able to:

a. Assess career path in IT.

1.1 Introduction

We live in a time of great change career-wise; jobs are lost to automation and others are changing what skills are required. With changes like these in the market, you may be considering a shift to a career in technology or, perhaps, you're already in the field and want to advance. In many industries, there continues to be strong demand for highly skilled IT specialists.

1.2 Computer Careers and Certification

The Computer Industry

- A demand for computer professionals continues to grow
- IT is predicted to be the fastest growing industry for the next several years

Careers in the Computer Industry

Job opportunities in the computer industry generally are available in one or more of these areas:

1. General business and government organizations and their IT departments
2. Computer equipment field
3. Computer software field
4. Computer service and repair field
5. Computer sales
6. Computer education and training field
7. IT consulting

- Employees in the **IT department** work together as a team to meet the information requirements of their organization
- Employees also are responsible for keeping all the computer operations and networks running

Jobs in the IT department typically are divided into six main areas:

1. Management
2. System development and programming
3. Technical services
4. Operations
5. Training
6. Security

Top 10 Highest Paying IT Jobs in 2020

Here's a look at just some of the highest-paying IT jobs, according to Robert Half Technology's 2020 Salary Guide and Simplilearn:

1. Big data engineer

Businesses need individuals who can transform large amounts of raw data into actionable information for strategy-setting, decision-making, and innovation — and pay well for people with these skills. The salary midpoint (or median national salary) for big data engineers is $163,250. These professionals typically create a company's software and hardware architecture, along with the systems people need to work with the data. Big data engineers usually have a degree in computer science and expertise in mathematics and databases.

2. Mobile applications developer

Just look at your phone or tablet applications and it's pretty easy to figure out why mobile applications developers are in demand. These IT pros need expertise in developing applications for popular platforms, such as iOS and Android. They also must have experience coding with mobile frameworks and mobile development languages, as well as a knowledge of web development languages. The salary midpoint for mobile application developers is $146,500.

3. Information systems security manager

Successful candidates for this hot job possess a technical background in systems and network security, but also have great interpersonal and leadership abilities. Analytical and problem-solving skills are key, as are excellent communication abilities. Information systems security managers also need to keep up with security trends and government regulations. Certifications such as Certified Information Systems Security Professional (CISSP) or CompTIA Security+ are often requested by employers. These IT pros earn a midpoint salary of $143,250.

4. Applications architect

These tech pros, who have a mean salary of $141,750, design the main parts of applications, including the user interface, middleware, and infrastructure. In addition to strong technical abilities, they need to be able to work well on teams and sometimes manage them. Excellent communication and planning skills are required for this job. It's one of the highest-paying IT jobs because just about every company wants to improve existing applications or create new ones.

5. Data architect

These tech professionals are responsible for the complicated processes essential to making strategic business decisions. They translate business requirements into database solutions and oversee data storage (data centers) and how the data is organized. Ensuring the security of those databases is part of the job as well. The salary midpoint for data architects is $141,250.

6. Database manager

Just about every business has some sort of database, and these tech pros maintain and support a firm's database environment, helping companies use data more strategically to meet their goals. And they get a midpoint salary of $133,500 to do it. Database managers are especially needed in large organizations with voluminous amounts of data to manage and must possess strong leadership and strategic planning skills.

7. Internet of Things (IoT) Solutions Architect

The IoT solutions architect is a leadership role in overseeing the strategy behind the development and deployment of IoT solutions. In addition to understanding IoT solutions, one should also have strong programming skills, an understanding of

Machine Learning, and knowledge of hardware design and architecture. An IoT solutions architect is responsible for leading as well as participating in the activities around architecture and design, helping to develop an overall IoT ecosystem engagement based on the IoT Solution Framework, and translating business needs into solution architecture requirements.The average yearly salary of an IoT solutions architect is $133,000.

8. Blockchain Engineer

A blockchain engineer specializes in developing and implementing architecture and solutions using blockchain technology. Due to the rapid rise of this technology, we already have a shortage of trained professionals for this role.

A blockchain engineer should have solid programming skills and a thorough understanding of the technologies behind Ripple, R3, Etherium, and Bitcoin as well as consensus methodologies and the security protocol stacks, crypto libraries, and functions. The average yearly salary of a blockchain engineer is $130,000.

9. Data security analyst

Data security analysts must thoroughly understand computer and network security, including firewall administration, encryption technologies, and network protocols. The job also requires excellent communication and problem-solving skills and knowledge of trends in security and government regulations. A professional certification, such as a Certified Information Systems Security Professional (CISSP) designation, is beneficial. The salary midpoint for data security analysts is $129,000.

10. Software engineer

The mean national salary for software engineers is $125,750. They design and create engineering specs for both applications and software, which means they're almost always in demand. Software engineers must have information systems knowledge and typically a bachelor's degree in computer science or a related area. Specific programming language knowledge is required, as well as strong communication skills.

Although these are some of the highest paying jobs in technology, plenty of other fields like cybersecurity and digital marketing are short on skilled professionals so keep looking if none of the careers described above appeal to you. If one of them does sound like a domain you'd like to transition into, keep in mind that you have various options such as going to school or pursue certifications through online learning and begin the process of training for one of these new roles and follow the money into one of these highest paying jobs.

General business and government organizations and their IT departments

Computer Equipment Field

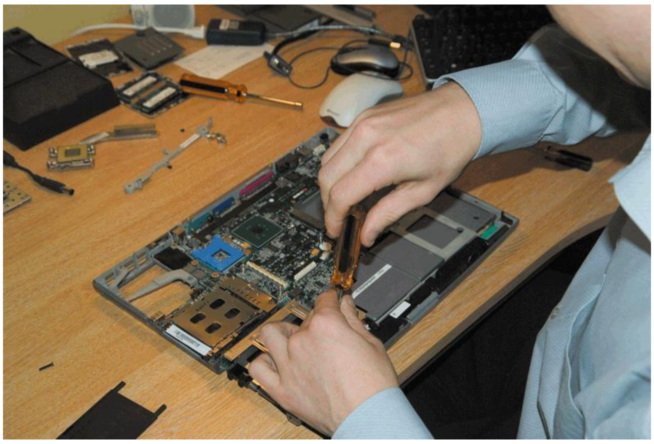
- The **computer equipment field** consists of manufacturers and distributors of computers and computer-related hardware
- Careers in this field are available with companies that design, manufacture, and produce computers and devices

Computer Software Field

The computer software field consists of companies that develop, manufacture, and support a wide range of software. Job titles might include:

1. Project leader
2. Project manager
3. Desktop or mobile application programmer/developer
4. Technical lead
5. Software engineer
6. Computer scientist

Computer Service and Repair Field



The **computer service and repair field** requires a knowledge of electronics provides:

1. Preventive maintenance
2. Component installation
3. Repair services

Computer Salespeople

Computer salespeople must possess a general understanding of computers and a specific knowledge of the products they are selling. Some work for computer equipment and software manufacturers, and others work for retailers.

Corporate Trainers

Corporate trainers teach employees how to use software, design and develop systems, write programs, and perform other computer-related activities.

1.2.1 Preparing for a Career in the Computer Industry

To prepare for a career in the computer industry, you first must decide on the area in which you are interested and then become educated in that field. If you desire a formal education, several options are available, which include attending a trade school, a college that offers two-year degrees, or a college or university that offers four-year degrees. Some classes are held on campus; others are offered online. After obtaining your education, you must remain current with changes in the field. The following sections discuss various options for obtaining formal computer education and methods of remaining current after embarking on a career in the computer industry.

Attending a Trade School

A trade school, also called a technical school, vocational school, or career college, offers programs primarily in the areas:

1. Programming
2. Web design and development
3. Graphics design
4. Hardware maintenance
5. Networking
6. Personal computer support
7. Security

If you attend a two-year college, check for an articulation agreement with a four-year university. Three broad disciplines produce the greatest number of entry-level employees.

Computer Discipline Differences

Computer Information Systems*	Computer Science	Computer Engineering
Practical and application oriented	Theory oriented	Design oriented
Business and management oriented	Mathematics and science oriented	Mathematics and science oriented
Understand how to design and implement information systems	Understand the fundamental nature of hardware and software	Understand the fundamental nature of hardware and electronics
Certificates Degrees include A.A., A.A.S., A.S., B.A., B.S., M.S., Ph.D.	Degrees include B.S., M.S., Ph.D.	Degrees include B.S., M.S., Ph.D.

*Sometimes called Information Technology or Management Information Systems

Attending a College or University

At colleges and universities, three broad disciplines produce the majority of entry-level employees in the computer industry: computer information systems, computer science, and computer engineering. Another program sometimes offered is software engineering, whose definition varies depending on the school, which may combine characteristics from each of these disciplines. The characteristics of each program are summarized in the following figure:

Computer information systems

- Programming
- Operating systems
- Systems analysis and design
- Databases
- Networking
- Web development

Computer Science

- Artificial intelligence
- Data communications
- Data and file structures
- Operating systems
- Application and systems programming
- Systems analysis and design

Computer Engineering

- Application and systems programming
- Data and file structures
- Networks
- Operating systems

Computer professionals with common interests and a desire to extend their proficiency form computer-related professional organizations.

A user group is a collection of people with common computer equipment or software interests. It can be an effective and rewarding way to learn about and continue career development.

1.3 Certification & Guide to Certification

Information technology is a highly dynamic and ever-changing field. As the industry evolves, so do the required skills needed for strong employees. In order to stay up-to-date with these new technologies, as well as competitive in general, it is important to regularly continue your education. One of the best ways to do this is by getting an IT certification.

Certification is the process of verifying the technical knowledge of an individual who has demonstrated competence in a particular area. Computer certifications are available in these areas:

1. Application software
2. Operating systems
3. Programming
4. Hardware
5. Networking
6. Digital forensics
7. Security

8. The Internet
9. Database systems

Selecting a certification requires careful thought and research.

Factors to Consider in Selecting a Certification

- Consider the expenses and the time involved to obtain the certification, including how often it must be renewed
- Examine employment projections
- Look at job listings to see what certifications are sought
- Read evaluations of certifications
- Talk to people in the industry
- Think about complementary combinations of certifications to meet your goals

Certification training options are available to suit every learning style.

Self-study	Online training classes	Instructor-led training	Web resources
• Requires high motivation • Least expensive	• Students set their own pace • Can be one-third the price of instructor-led programs	• Typically span from several days to several months • Some sponsors hold their own sessions	• May include detailed course objectives, training guides, sample test questions, chat rooms, and discussion groups

Certifications are typically taken on a computer at a testing center. With computerized adaptive testing (CAT), the tests analyze a person's responses while taking the test.

Once you choose the certification you want to earn, it's time to prepare.

- Learn all about the certification process – for example, where and how will you take the test, how much will it cost, how do you register, are there prerequisites and will you need any special supplies or equipment.
- Choose your training option. Will you take a class, use a self-paced e-learning tool or study on your own? Decide which makes the most sense for you based on your learning style and timeline.
- Become familiar with the exam by downloading official exam objectives and sample questions and by reading what other IT pros have said about the exams.
- Register and pay for your exam and mark your calendar. Give yourself ample time to study so you can go in confidently and ace your test.

Why should I consider IT Certification?

According to CompTIA, the largest vendor-neutral certifying group, 91% of employers believe IT certifications play a key role in the hiring process and that IT certifications are a reliable predictor of a successful employee.

Stay Relevant

Continuing education is important in any industry, but it's crucial in a field like technology. IT certifications help professionals stay relevant in their specialty, requiring you to attend conferences, participate in webinars, take classes, or write for publications sharing your subject matter expertise.

Ramp Up Your Skills

Whether you are looking for a new position entirely, or are looking at moving up through promotions, many people apply for jobs that allow them to further their growth, typically without having the necessary skills required at the time of hire. By obtaining an IT certification through classes and online training modules, it allows you to gain the skills needed in order to meet your career goals. This knowledge can be immediately applied on the job and provides instant advantages.

Invest in Yourself

New knowledge and education are never wasted. By getting a certification, it not only shows to your employer (current or future) that you are invested in your career, but it also proves that you are a lifelong learner that never wants to stop improving. Obtaining an IT certification only furthers your knowledge and could move you in a different direction that you are passionate about.

What are the top IT certifications?

Global Knowledge recently published the 15 top-paying IT certifications in 2019 based on the results of their 2019 IT Skills & Salary Report research. These reveal the current philosophy shifts and tech trends impacting IT departments in the United States, with most emphasizing cloud computing, project management, cybersecurity, and networking.

1. **Google Certified Professional Cloud Architect**

This certification allows IT professionals to certify as a cloud architect on the Google Cloud Platform. It demonstrates the ability to design, develop and manage a secure, scalable and reliable cloud architecture using GCP technologies.

2. PMP – Project Management Professional

PMP was created and is administered by the Project Management Institute (PMI®) and is considered the most important industry-recognized certification for project managers. This certification provides assurance that the holder has both the content knowledge and field experience to effectively define, plan and deliver their projects.

3. Certified ScrumMaster

Created and managed by the Scrum Alliance, obtaining a Certified ScrumMaster certificate proves that you are well-verses in both understanding the Agile Scrum methodology as well as putting it into practice. Typically, a Certified ScrumMaster will lead a team and help them learn the Scrum framework while performing to the best of their abilities. Attaining this certification gives you a two-year membership to the Scrum Alliance.

4. AWS Certified Solutions Architect – Associate

The AWS Certified Solutions Architect Associate-level exam demonstrates an individual's expertise in designing and deploying scalable systems on AWS. With further market need for skilled and certified AWS solutions architects as cloud continues to grow, this certification will also continue to be in-demand.

5. AWS Certified Developer – Associate

The AWS Certified Developer – Associate certification validates technical expertise in developing and maintaining applications on AWS as well as being able to efficiently use AWS SDKs to interact with services from within applications and write code that optimizes AWS application performance.

6. Microsoft Certified Solutions Expert (MCSE): Core Infrastructure

This certification validates that you have the skills needed to run a highly efficient and modern data center, identity management, systems management, virtualization, storage, and networking. Earning an MCSE: Core Infrastructure certification qualifies you for such jobs as administrator, architect, computer support specialist, and information security analyst.

7. ITIL® Foundation

ITIL Foundation is the entry-level ITIL certification and provides a broad-based understanding of the IT service lifecycle. The newest version, ITIL 4, was released this year and introduces new ways of working, such as DevOps, Agile and Lean IT.

8. CISM – Certified Information Security Manager

ISACA created and maintains the CISM certification which is aimed at professionals who manage organizations' IT security. This certification requires at least five years of hands-on experience in information security, with at least three of those as a security manager.

9. CRISC – Certified in Risk and Information Systems Control

ISACA also offers and manages this certification which focuses on risk management proficiency. CRISC certification proves professionals gave the ability to help organizations understand their business risk as well as develop, implement, and maintain information systems controls.

10. CISSP – Certified Information Systems Security Professional

Offered by the International Information Systems Security Certification Consortium $(ISC)^2$ as a vendor-neutral credential, CISSP is designed to prove security expertise, and is one of the most sought-after certifications in cybersecurity. The exam covers a variety of domains, including areas such as security and risk management, communications and network security, software development security, asset security, security architecture and engineering, identity and access management, security assessment and testing, and security operations.

11. CEH – Certified Ethical Hacker

The International Council of E-Commerce Consultants (EC-Council) created and manages the Certified Ethical Hacker certification, whose goal is to demonstrate ethical hacking methodologies that can be used in penetration testing. Passing the CEH exam is the only step needed to certification. The exam is designed to test a candidate's abilities to find weaknesses and vulnerabilities in a company's network defenses using techniques and methods that hackers typically employ.

12. Citrix Certified Associate – Virtualization (CCA-V)

The Citrix Certified Associate – Virtualization certification is entry-level and demonstrates the basics of managing, maintaining, monitoring and troubleshooting. In order to gain certification, hands-on experience and passing the exam is required.

13. CompTIA Security+

CompTIA Security+ is an intermediate-level certification that demonstrates mastery of security topics like data, application, host, network, physical and operational security.

14. CompTIA Network+

The CompTIA Network+ focuses on foundational networking skills, including installation, maintenance and troubleshooting of networks, protocols for both the LAN and WAN, and the importance of network security. Passing this exam proves that you have knowledge related to network infrastructure and network protocols. The CompTIA IT Fundamentals certification and CompTIA A+ certification are prerequisites for this certification.

15. Cisco Certified Networking Professional (CCNP) Routing and Switching

The Cisco Certified Networking Professional (CCNP) Routing and Switching certification is an advanced certification that ensures network engineers and administrators have the skills to plan, implement, verify and troubleshoot local and wide area enterprise networks. It also shows that these professionals are able to work collaboratively with specialists on advanced security, voice, wireless and video solutions.

ASSESMENT

Direction:

Assess yourself if what IT career opportunities would you like to pursue. Discuss briefly your reasons why it interests you. Please submit your answer in a form of video/vlog.

The following table shows the rubric that will be used in evaluating your video presentation/vlog.

Criteria	**Excellent** **5**	**Good** **4**	**Average** **3**	**Needs improvement** **2**
Organization	The presentation was very easy to follow	The presentation was easy to follow.	. The presentation was not easy to follow.	Organization The presentation was difficult to follow due to disorganization of the utterances.
Accuracy of language	The student communicated well using correct vocabulary and grammar	The student made a few mistakes in vocabulary and grammar but there were no patterns of errors.	The student made some mistakes in vocabulary and grammar.	Accuracy of language use It was hard to understand due to incorrect use of vocabulary and grammar.
Understanding of topic.	The student clearly understood the topic in-depth and presented his/her information convincingly.	The student seemed to understand the main points of the topic and presented those with ease.	The student clearly understood most aspects of the topic and presented his/her information with ease.	The student did not show an adequate understanding of the topic.
Accuracy of information	All information presented in writing was clear, accurate and thorough.	Most information presented in writing was clear, accurate and thorough.	Most information presented in writing was clear, but was not usually accurate.	Most of the information was inaccurate or not clear.
Comprehension	The student was able to accurately answer almost all questions about the topic.	The student was able to accurately answer most questions about the topic	The student was able to accurately answer a few questions about the topic.	The student was unable to accurately answer questions about the topic.

1.3 References

Vermaat, Misty E. et al. Discovering Computers: Technology, Data, and Devices. Cengage Learning : USA, 2018.

https://www.informatics.edu.ph/2020/05/27/top-18-highest-paying-it-jobs-in-2020/

https://www.bmc.com/blogs/it-certifications/

https://books.google.com.ph/books/about/Discovering_Computers_Essentials_2018_Di.html?id=DAAxDgAAQBAJ&printsec=frontcover&source=kp_read_button&redir_esc=y#v=onepage&q&f=false

1.4 Acknowledgement

All the figures and information presented in this module were taken from the references enumerated above.

Unit 2
Digital Literacy: Introducing a World of Technology

2.0 Learning Outcomes

After completing this chapter, you will be able to:

a. Demonstrate steps on how to protect computer from malware and viruses.
b. Create a case report on information technology best practices in various applications in IT industry.

2.1 Introduction

Technology is an integral part of life at school, home, and work. Technology can enable you to more efficiently and effectively access and search for information; share personal ideas, photos, and videos with friends, family, and others; communicate with and meet other people; manage finances; shop for goods and services; play games or access other sources of entertainment; keep your life and activities organized; and complete business activities. Because technology changes, you must keep up with the changes to remain digitally literate. Digital literacy (Having a current knowledge and understanding of computers, mobile devices, the Internet, and related technologies.) involves having a current knowledge and understanding of computers, mobile devices, the Internet, and related technologies.

2.2 Computers

Computer is an electronic device, operating under the control of instructions stored in its own memory, that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use.

Characteristics of a Computer

1. **Speed –** a computer can perform tasks very fast, no human being can compete to solving the complex computation faster than computer.
2. **Accuracy –** since the computer is programmed, so whatever input we give it gives result accurately.
3. **Diligence –** computer can work for hours without any break and creating error.
4. **Versatility –** computer can be used to perform completely different type of work at the same time.
5. **Power of remembering –** every piece of information that a user stores on a computer can be retained as long as is needed.
6. **No feeling –** computers are devoid of emotion, they have no feeling and instincts because they are machine.

2.2.1 Mobile and Game Devices

Mobile Device

Mobile device is a computing device small enough to hold in your hand.

Are mobile devices computer?

The mobile devices discussed in this section can be categorized as computers because they operate under the control of instructions stored in their own memory, can accept data, process the data according to specified rules, produce or display information, and store the information for future use.

Types of Mobile Devices

1. **Smartphone** - is an Internet-capable phone that usually also includes a calendar, an appointment book, an address book, a calculator, a notepad, games, and several other apps (which are programs on a smartphone).
2. **Digital camera -** is a mobile device that allows you to take photos and store the photographed images digitally.
3. **Portable media player** - mobile device on which you can store, organize, and play or view digital media and sometimes called a personal media player.

Game Devices

Game device is a mobile computing device designed for single-player or multiplayer video games. Gamers often connect the game console to a television so that they can view their gameplay on the television's screen

Games Accessories and Input Techniques

1. **Gamepad -** holding the gamepad with both hands, press buttons with your thumbs or move sticks in various directions to trigger events.
2. **Air gesture** - moving your body or a handheld device in predetermined directions through the air, you may need at least six feet of open space to accommodate your motions.
3. **Voice command** - speaking instructions toward the game console, ensure no background noise is in the room, or use a headset.
4. **Fitness Accessories -** working with fitness accessories, such as balance boards and resistance bands, you can make fitness fun and functional.

Categories of Games

1. **Action and adventure:** Characters move through their world by running, jumping, and climbing in an attempt to reach the next level of play and eventually defeat a villain or rescue a prisoner.

2. **Education:** Engaging and entertaining puzzles and problems present practical instruction and possible assessment to teach skills and concepts that are applicable in real-world circumstances.
3. **Music and fitness:** Individuals and groups improve their physical and mental fitness while engaging in athletic competitions, aerobics, dance, and yoga.
4. **Puzzle and Strategy:** Players use their skills and intelligence to solve problems, generally with the goal of moving objects to create a pattern.
5. **Racing and Sports:** Athletes and drivers strive to complete a course or to cross a finish line in record time.
6. **Role-playing:** Gamers assume the role of a character and experience adventures while on a major quest.
7. **Simulation:** Players control an activity in a simulated situation, such as piloting an airplane or playing an instrument in a rock band.

2.2.2 Data and Information

Data is a collection of unprocessed items, which can include text, numbers, images, audio, and video.

Information is a processed data that conveys meaning to users.) conveys meaning to users. Both business and home users can make well-informed decisions because they have instant access to information from anywhere in the world.

Data Processing Activities

1. **Collection** - data originates in the form of events transaction or some observations. This data is then recorded in some usable form.
2. **Conversion** - once the data is collected, it is converted from its source documents to a form that is more suitable for processing.
3. **Manipulation** - once data is collected and converted, it is ready for the manipulation function which converts data into information. Manipulation consists of following activities:
   a. **Sorting** - it involves the arrangement of data items in a desired sequence.
   b. **Calculating -** arithmetic manipulation of data is called calculating. Items of recorded data can be added to one another, subtracted, divided or multiplied to create new data. Calculation is an integral part of data processing.

c. **Summarizing - t**o summarize is to condense or reduce masses of data to a more usable and concise form.

d. **Comparing - t**o compare data is to perform an evaluation in relation to some known measure.

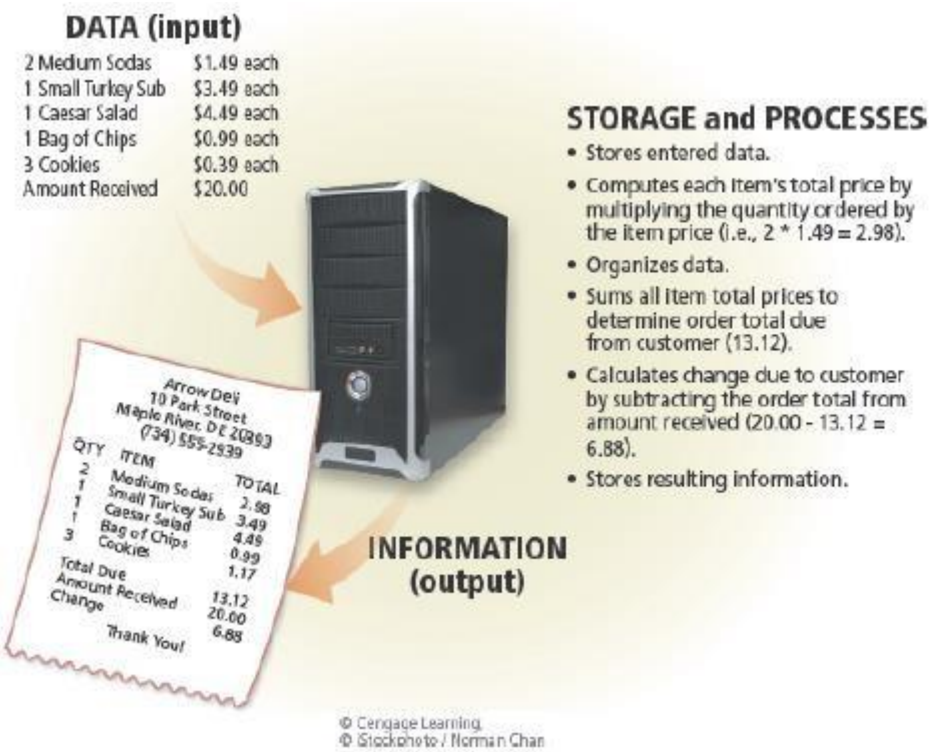
4. **Managing the Output Results - o**nce data has been captured and manipulated following activities may be carried out:
   a. **Storing -** to store is to hold data for continued or later use. Storage is essential for any organized method of processing and re-using data. The storage mechanisms for data processing systems are file cabinets in a manual system,

and electronic devices such as magnetic disks/magnetic tapes in case of computer based system.

b. **Retrieving -** to retrieve means to recover or find again the stored data or information. Retrieval techniques use data storage devices. Thus data, whether in file cabinets or in computers can be recalled for further processing. Retrieval and comparison of old data gives meaning to current information.

5. **Communication** - is the process of sharing information. Unless the information is made available to the users who need it, it is worthless. Thus, communication involves the transfer of data and information produced by the data processing system to the prospective users of such information or to another data processing system.
6. **Reproduction -** to reproduce is to copy or duplicate data or information. This reproduction activity may be done by hand or by machine.

These data processing activities is shown in the following figure.



2.2.3 Internet

Internet is a worldwide collection of computer networks that connects millions of businesses, government agencies, educational institutions, and individuals.

World Wide Web (or web, for short) is a global library of information available to anyone connected to the Internet. People around the world access the web to accomplish a variety of online tasks, including:

- Search for information

- Conduct research
- Communicate with and meet other people
- Share information, photos, and videos with others
- Access news, weather, and sports
- Participate in online training
- Shop for goods and services
- Play games with others
- Download or listen to music
- Watch videos
- Download or read books

Website is a collection of related webpages, which are stored on a web server.

Web server is a computer that delivers requested webpages to your computer or mobile device.

2.2.4 Digital Security and Safety

People rely on computers to create, store, and manage their information. To safeguard this information, it is important that users protect their computers and mobile devices.

Malware is short for malicious software, is software that acts without a user's knowledge and deliberately alters the computer's and mobile device's operations. Examples of malware include viruses, worms, trojan horses, rootkits, spyware, adware, and zombies.

Each of these types of malware attacks your computer or mobile device differently. Some are harmless pranks that temporarily freeze, play sounds, or display messages on your computer or mobile device. Others destroy or corrupt data, instructions, and information stored on the infected computer or mobile device.

Protection from Viruses and other Malware

1. **Use virus protection software.** Install a reputable antivirus program and then scan the entire computer to be certain it is free of viruses and other malware. Update the antivirus program and the virus signatures (known specific patterns of viruses) regularly.
2. **Use a firewall.** Set up a hardware firewall or install a software firewall that protects your network's resources from outside intrusions.
3. **Be suspicious of all unsolicited email attachments.** Never open an email attachment unless you are expecting it and it is from a trusted source. When in doubt, ask the sender to confirm the attachment is legitimate before you open it. Delete or quarantine flagged attachments immediately.

4. **Disconnects our computer from the Internet**. If you do not need Internet access, disconnect the computer from the Internet. Some security experts recommend disconnecting from the computer network before opening email attachments.
5. **Download software with caution.** Download programs or apps only from websites you trust, especially those with music and movie sharing software.
6. **Close spyware windows.** If you suspect a pop-up window (rectangular Internet Research area that suddenly appears on your screen) may be spyware, close the window. Never click an Agree or OK button in a suspicious window.
7. **Before using any removable media, scan it for malware.** Follow this procedure even for shrink-wrapped software from major developers. Some commercial software has been infected and distributed to unsuspecting users. Never start a computer with removable media inserted in the computer unless you are certain the media are uninfected.
8. **Keep current.** Install the latest updates for your computer software. Stay informed about new virus alerts and virus hoaxes.
9. **Back up regularly.** In the event your computer becomes unusable due to a virus attack or other malware, you will be able to restore operations if you have a clean (uninfected) backup.

2.2.5 Programs and Applications

Programs consists of a series of related instructions, organized for a common purpose, that tells the computer what tasks to perform and how to perform them.

Two Categories of Software

1. **System Software –** consists of the program that control or maintain the operations of the computer and its devices.

Example:

Operating system - is a set of programs that controls the computer hardware and acts as an interface with applications. Operating systems can control one or more computers, or they can allow multiple users to interact with one computer.

Utility programs help to perform maintenance or correct problems with a computer system.

2. **Application Software** – consists of programs designed to make users more productive and/or assist them with personal tasks.

Categories of Applications and Its Uses

Category	Sample Application	Sample Uses
Productivity	Word Processing Presentation Schedule and Contact Management Personal Finance	Create letters, reports, and other documents. Create visual aids for presentations. Organize appointments and contact lists. Balance checkbook, pay bills, and track income and expenses.
Graphics and Media	Photo Editing Video and Audio Editing Media Player	Modify digital photos, i.e., crop, remove red-eye, etc. Modify recorded movie clips, add music, etc. View images, listen to audio/music, watch videos.
Personal Interest	Travel, Mapping, and Navigation Reference Educational Entertainment	View maps, obtain route directions, locate points of interest. Look up material in dictionaries, encyclopedias, etc. Learn through tutors and prepare for tests. Receive entertainment news alerts, check movie times and reviews, play games.
Communications	Browser Email VoIP FTP	Access and view webpages. Send and receive messages. Speak to other users over the Internet. Transfer items to and from other computers on the Internet.

Security	Antivirus Personal Firewall Spyware, Adware, and other Malware Removers	Protect a computer against viruses. Detect and protect against unauthorized intrusions. Detect and delete spyware, adware, and other malware.

Desktop Applications are those applications stored on a computer. Often is used to describe applications stored on a computer.

Web Applications are those applications stored on a web server that you access through a browser.

Mobile Applications are those applications you download from a mobile device's application store or other location.

2.2.6 Communications and Networks

Computer communications describe a process in which two or more computers or devices transfer (send and receive) data, instructions, and information over transmission media via a communications device(s).

Communications Device - is hardware capable of transferring items from computers and devices to transmission media and vice versa. Examples of communications devices are modems, wireless access points, and routers.

Network is a collection of computers and devices connected together, often wirelessly, via communications devices and
transmission media.

Types of Networks

1. **Local Area Network (LAN) -** refers to a group of computers that all belong to the same organization and that are linked within a small geographic area using a network and often the same technology.

2. **Metropolitan Area Network (MAN) -** connects multiple geographically close LANs (over an area of up to several dozen miles) to one another at high speeds. Thus, a MAN lets two remote nodes communicate as if they were part of the same local area network.

3. **Wide Area Network (WAN) -** connects multiple LANs to one another over great geographic distances. The speed available on a WAN varies depending on the cost of the connections, which increases with distance, and may be low.

2.2.7 Netiquette

Netiquette is the code of acceptable Internet behavior.

Netiquette Guidelines for Online Communications **Golden Rule: Treat others as you would like them to treat you.**
Be polite. Avoid offensive language.
Avoid sending or posting *flames*, which are abusive or insulting messages. Do not participate in *flame wars*, which are exchanges of flames.
Be careful when using sarcasm and humor, as it might be misinterpreted.
Do not use all capital letters, which is the equivalent of SHOUTING!
Use **emoticons** to express emotion. Popular emoticons include: :) Smile :| Indifference :o Surprised :(Frown :\ Undecided ;) Wink
Use abbreviations and acronyms for phrases: BTW by the way IMHO in my humble opinion FWIW for what it's worth FYI for your information TTFN ta ta for now TYVM thank you very much
Clearly identify a spoiler, which is a message that reveals an outcome to a game or ending to a movie or program.
Be forgiving of other's mistakes.
Read the *FAQ* (frequently asked questions), if one exists.

2.2.8 The Core Rules of Netiquette — Summary

Rule 1. Remember the human.
Never forget that the person reading your mail or posting is, indeed, a person, with feelings that can be hurt.
Corollary 1 to Rule #1: It's not nice to hurt other people's feelings.
Corollary 2: Never mail or post anything you wouldn't say to your reader's face.
Corollary 3: Notify your readers when flaming.

Rule 2. Adhere to the same standards of behavior online that you follow in real life.
Corollary 1: Be ethical.
Corollary 2: Breaking the law is bad Netiquette.

Rule 3. Know where you are in cyberspace.

Corollary 1: Netiquette varies from domain to domain.
Corollary 2: Lurk before you leap.

***Rule 4.* Respect other people's time and bandwidth.**

Corollary 1: It's OK to think that what you're doing at the moment is the most important thing in the universe, but don't expect anyone else to agree with you.
Corollary 2: Post messages to the appropriate discussion group.
Corollary 3: Try not to ask stupid questions on discussion groups.
Corollary 4: Read the FAQ (Frequently Asked Questions) document.
Corollary 5: When appropriate, use private email instead of posting to the group.
Corollary 6: Don't post subscribe, unsubscribe, or FAQ requests.
Corollary 7: Don't waste expert readers' time by posting basic information.

ASSESMENT

Direction: Below are the tasks you need to follow.

Task 1: Create a video presentation/vlog, showing that you are demonstrating the steps on how to protect computer or mobile devices from malware and other viruses. Also, you need to perform the core rules of netiquette.

Task 2: Create a case report on information technology best practices in various applications in IT industry.

The following table shows the rubric that will be used by your instructor in evaluating your case study report.

	Excellent 5	**Good** 3	**Average** 2	**Needs improvement** 1
Knowledge and argumentation	Strong use of sources - including summary, quotations, interpretation and analysis. Argumentation.	Good use of sources and some argumentation.	Some use of sources and very little or unclear argumentation.	Insufficient use of sources and/or sources misunderstood. Unclear or lacking argumentation.
Vocabulary and style	Wide and growing vocabulary. Variety of sentence types.	Most elements achieved.	Some elements achieved.	Vocabulary in this paper is limited. There are several instances of incorrect word choices. There is a narrow range of sentence variety.
Paragraphs and punctuation	Main ideas appropriately divided by paragraphs. Sentences divided by proper punctuation.	Good control.	Paper has examples of proper use of paragraphs and punctuation.	Use of paragraphs and punctuation needs improvement.
Spelling and grammar	Precise and consistent control.	Good control. Paper has some errors that do not interfere with understanding.	Paper has several errors that could interfere with understanding or affect readability.	Several sentences and/or ideas are difficult to understand because of errors.

The following table shows the rubric that will be used by your instructor in evaluating your video presentation/vlog.

Criteria	Excellent 5	Good 4	Average 3	Needs improvement 2
Organization	The presentation was very easy to follow	The presentation was easy to follow.	. The presentation was not easy to follow.	Organization The presentation was difficult to follow due to disorganization of the utterances.
Accuracy of language	The student communicated well using correct vocabulary and grammar	The student made a few mistakes in vocabulary and grammar but there were no patterns of errors.	The student made some mistakes in vocabulary and grammar.	Accuracy of language use It was hard to understand due to incorrect use of vocabulary and grammar.
Understanding of topic.	The student clearly understood the topic in-depth and presented his/her information convincingly.	The student seemed to understand the main points of the topic and presented those with ease.	The student clearly understood most aspects of the topic and presented his/her information with ease.	The student did not show an adequate understanding of the topic.
Accuracy of information	All information presented in writing was clear, accurate and thorough.	Most information presented in writing was clear, accurate and thorough.	Most information presented in writing was clear, but was not usually accurate.	Most of the information was inaccurate or not clear.
Comprehension	The student was able to accurately answer almost all questions about the topic.	The student was able to accurately answer most questions about the topic	The student was able to accurately answer a few questions about the topic.	The student was unable to accurately answer questions about the topic.

2.3 References

Vermaat, Misty E. et al. Discovering Computers: Technology, Data, and Devices. Cengage Learning : USA, 2018.

https://books.google.com.ph/books/about/Discovering_Computers_Essentials_2018_Di.html?id=DAAxDgAAQBAJ&printsec=frontcover&source=kp_read_button&redir_esc=y#v=onepage&q&f=false

http://www.albion.com/netiquette/book/0963702513p32.html

2.4 Acknowledgement

All the figures and information presented in this module were taken from the references enumerated above.